

IC 7660 (IC1) is a switched-capacitor voltage converter that produces a negative output voltage equal to the inverse of its positive supply voltage. The amount of current that it is able to supply is limited, so the negative voltage does not match the positive supply. In this circuit, it is about -8V, which is adequate for peak input voltage of up to about 7V.

Assemble the circuit on a general-purpose PCB and enclose in a suitable case. Use crocodile clips as input connectors. These can be clipped to an appropriate point in the test circuit. The output leads are

terminated in 4mm banana plugs that are to be plugged into the terminals of a multimeter in place of the casual test probe.

To test the circuit, connect a 10-kilo-ohm or 100-kilo-ohm potentiometer across the 9V supply. Connect the wiper of the potentiometer to the input of the circuit. By turning the knob of the potentiometer, you can deliver to the input a voltage varying from 0V to 9V. Start with the wiper at 0V end of the track so that the circuit receives no voltage.

Connect a meter to the output and set it to the 10V or 20V scale. If you

have a second meter, you can connect it to the input to monitor the input voltage. Press S2 to reset the circuit and vary the potentiometer to increase the input voltage to, say, 2V. The output should read 2V. Now decrease the input voltage to 1V. The output voltage should still read nearly 2V, though it can be seen falling slowly. Press S2 to reset the input. The output will fall to 0V but instantly rise to 1V when the button is released. Repeat for a few other voltage levels ranging from 1V to 7V to confirm that everything is working correctly. ●

ELECTRONIC WATER ALARM

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A burst water-supply hose of the washing machine, a bathroom tap that you forgot to close, or a broken aquarium wall may turn your house into a pond. You can avoid this mess by using an electronic water alarm that warns you of the water leakage as soon as possible.

The acoustic water alarm circuit presented here takes advantage of the fact that the tap water is always slightly contaminated (or has salts and minerals) and thus conducts electricity to a certain extent. It is built around IC LMC555 (IC1), which is a CMOS version of the bipolar 555 timer chip.

IC1 is followed by a complementary pair of emitter followers (T1 and T2) to drive a standard 8-ohm speaker (LS1). Power is supplied by a compact 9V PP3 battery.

Power is applied when power switch S1 is closed. The reset input (pin 4) of IC1 is held low by resistor R1 (2.2-kilo-ohm). The astable oscillator wired around IC1 is in disabled mode. When probes P1 and P2 become wet, these conduct to reverse the state of IC1's reset terminal. As a result, the astable multivibrator starts oscillating at a frequency determined by resistor R2 and capacitor C3. The output of IC1 drives the complementary pair of transistors T1 and T2.



Although this combination causes significant crossover distortion, it doesn't have any adverse effect on the square-wave audio signal processing. A 10-kilo-ohm potentiometer (VR1) is inserted between output pin 3 of IC1 and the bases of transistors T1 and T2 for volume control.

The probes can be made using two suitable copper needles or small pieces of circuit board with the copper surface coated with solder. Fit these at the lowest point where water will accumulate. After construction, place the alarm circuit well away from the point of possible leakage. Use a pair of thin twisted flexible wires to connect the probes to the circuit.

Capacitor C1 connected across IC1 input (pin 4 and GND) keeps the alarm circuit from responding to stray electrostatic fields. Similarly, twisting the wires together makes the relatively long connection between the probes and the circuit less sensitive to false alarms due to external electromagnetic interference. Finally, if you want to lower the probe sensitivity, reduce the value of grounding resistor R1. ●

